

Trigonometric Functions & the Unit Circle



Radians, degrees, and the unit circle

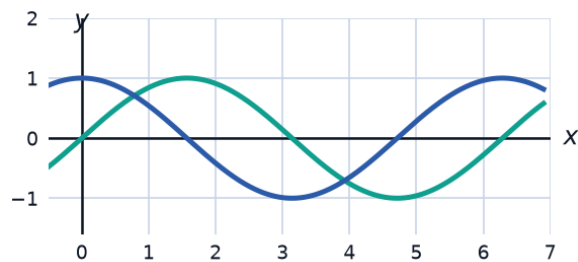
- 1 Convert 150° to radians, and convert $\frac{5\pi}{4}$ radians to degrees.
 - 2 Find the exact coordinates of the point on the unit circle at angle $\theta = \frac{2\pi}{3}$.
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Exact values from the unit circle

- 3 Find the exact values of $\sin\left(\frac{7\pi}{6}\right)$ and $\cos\left(\frac{7\pi}{6}\right)$.
 - 4 Find the exact value of $\tan\left(\frac{5\pi}{3}\right)$.
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Visualizing sine and cosine graphs

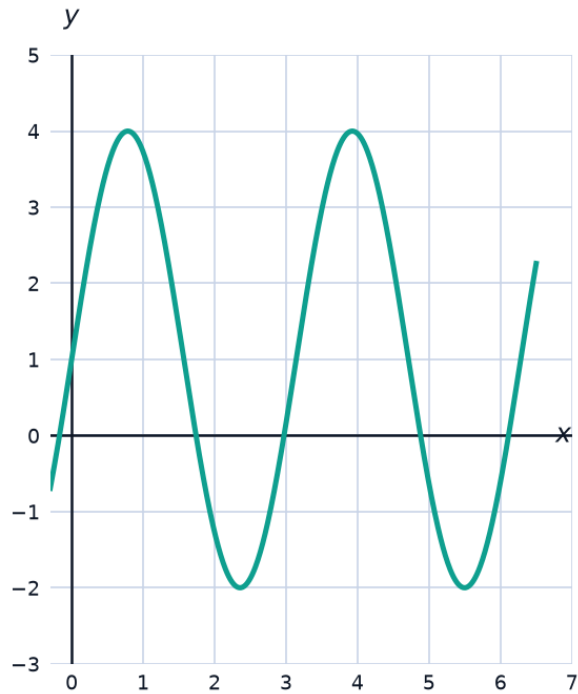
- 5 The graph shows $f(x) = \sin(x)$ and $g(x) = \cos(x)$ over one period.



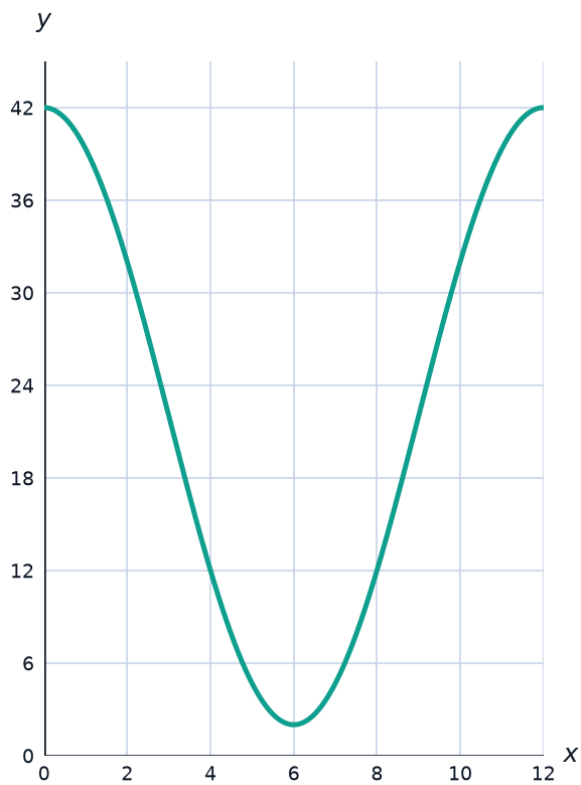
- a State the amplitude and period of each function shown.
 - b Identify the horizontal shift that maps $\sin(x)$ onto $\cos(x)$.
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Amplitude, period, and vertical shift

- 6 State the amplitude, period, and vertical shift of $f(x) = 3\sin(2x) + 1$.



- 7 A Ferris wheel's height above the ground is modeled by $h(t) = 20\cos\left(\frac{\pi}{6}t\right) + 22$, where t is in minutes. Find the maximum height, minimum height, and period of one full rotation.

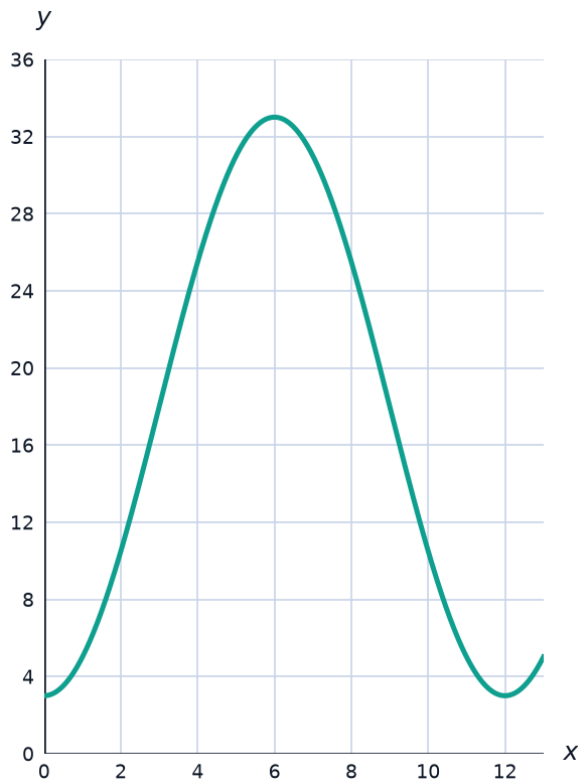


Reciprocal and Pythagorean identities

- 8 Given $\sin \theta = \frac{3}{5}$ with θ in Quadrant II, find $\cos \theta$ and $\tan \theta$.
- 9 Find $\sec \theta$ and $\csc \theta$ if $\cos \theta = -\frac{5}{13}$ and θ is in Quadrant III.
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Modeling periodic phenomena

- 10 The average daily temperature in a city is modeled by $T(m) = 15\sin\left(\frac{\pi}{6}(m-3)\right) + 18$ degrees Celsius, where m is the month number ($m = 1$ for January). Find the temperature in April ($m = 4$) and state the month of maximum temperature.



Co-function and even/odd identities

- 11 Use a co-function identity to rewrite $\cos\left(\frac{\pi}{5}\right)$ in terms of sine.
- 12 Use the fact that sine is an odd function and cosine is an even function to evaluate $\sin(-30^\circ)$ and $\cos(-60^\circ)$.
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Solving basic trigonometric equations

- 13 Solve $2\sin \theta - 1 = 0$ for θ in $[0, 2\pi)$.
- 14 Solve $2\cos^2 \theta - 1 = 0$ for θ in $[0, 2\pi)$.
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Angular and linear speed

- 15 A wheel of radius 10 cm rotates at 4 revolutions per second. Find its angular speed ω in radians per second, and the linear speed of a point on its rim.
- 16 A satellite orbits Earth at a radius of 7000 km, completing one orbit every 2 hours. Find its linear speed in km/h.
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Right-triangle definitions and applications

- 17 A ladder leans against a wall, making a 65° angle with the ground. If the ladder is 12 ft long, find the height it reaches on the wall, to one decimal place.
- 18 From a point 50 ft from the base of a tower, the angle of elevation to the top is 40° . Find the height of the tower, to one decimal place.
- 19 A kite string is 80 ft long and makes a 55° angle with the ground. Find the height of the kite above the ground, and the horizontal distance from the flyer to a point directly below the kite, both to one decimal place.
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Multiple choice — easy 15

- 20 Find the exact value of $\sin(0)$.

- a. $\frac{1}{2}$
- b. 0
- c. $\frac{\sqrt{2}}{2}$
- d. $-\frac{\sqrt{2}}{2}$

21 Find the exact value of $\sin\left(\frac{7\pi}{6}\right)$.

a. 0

b. $\frac{\sqrt{3}}{2}$

c. $-\frac{1}{2}$

d. 1

22 Find the exact value of $\sin\left(\frac{5\pi}{4}\right)$.

a. $-\frac{\sqrt{3}}{2}$

b. -1

c. $-\frac{\sqrt{2}}{2}$

d. $\frac{\sqrt{3}}{2}$

23 Find the exact value of $\cos\left(\frac{7\pi}{6}\right)$.

a. $-\frac{\sqrt{3}}{2}$

b. 0

c. -1

d. $-\frac{1}{2}$

24 Find the exact value of $\cos(0)$.

a. 1

b. $\frac{\sqrt{2}}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. $-\frac{1}{2}$

25 Find the exact value of $\sin\left(\frac{2\pi}{3}\right)$.

a. $\frac{\sqrt{3}}{2}$

b. $-\frac{1}{2}$

c. 0

d. -1

26 Find the exact value of $\cos\left(\frac{11\pi}{6}\right)$.

a. $\frac{1}{2}$

b. 0

c. $\frac{\sqrt{3}}{2}$

d. $\frac{\sqrt{2}}{2}$

27 Find the exact value of $\sin\left(\frac{5\pi}{6}\right)$.

a. $\frac{1}{2}$

b. $-\frac{\sqrt{2}}{2}$

c. -1

d. 0

28 Find the exact value of $\cos\left(\frac{3\pi}{4}\right)$.

a. 0

b. $\frac{1}{2}$

c. $-\frac{1}{2}$

d. $-\frac{\sqrt{2}}{2}$

29 Find the exact value of $\sin\left(\frac{\pi}{3}\right)$.

a. $\frac{1}{2}$

b. $\frac{\sqrt{3}}{2}$

c. 1

d. $\frac{\sqrt{2}}{2}$

30 Find the exact value of $\sin\left(\frac{5\pi}{3}\right)$.

a. $\frac{\sqrt{3}}{2}$

b. $-\frac{\sqrt{3}}{2}$

c. 1

d. -1

31 Find the exact value of $\cos\left(\frac{\pi}{4}\right)$.

a. $\frac{1}{2}$

b. $-\frac{1}{2}$

c. 0

d. $\frac{\sqrt{2}}{2}$

32 Find the exact value of $\sin\left(\frac{\pi}{4}\right)$.

a. $-\frac{\sqrt{3}}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{1}{2}$

d. 0

33 Find the exact value of $\cos\left(\frac{5\pi}{6}\right)$.

a. $-\frac{\sqrt{2}}{2}$

b. $-\frac{\sqrt{3}}{2}$

c. 0

d. -1

34 Find the exact value of $\cos\left(\frac{4\pi}{3}\right)$.

a. $-\frac{1}{2}$

b. -1

c. $\frac{\sqrt{2}}{2}$

d. 0

Multiple choice — medium 15

35 Find the period of $f(x) = 3\sin(6x) - 4$.

a. $\frac{2\pi}{3}$

b. $\frac{\pi}{6}$

c. $\frac{\pi}{3}$

d. $\frac{4\pi}{3}$

36 Find the period of $f(x) = 8\sin(5x) + 7$.

a. $\frac{4\pi}{5}$

b. $\frac{\pi}{5}$

c. $\frac{7\pi}{5}$

d. $\frac{2\pi}{5}$

37 Find the period of $f(x) = 9\cos(7x) - 8$.

a. $\frac{4\pi}{7}$

b. $\frac{\pi}{7}$

c. $\frac{2\pi}{7}$

d. $\frac{9\pi}{7}$

38 Find the period of $f(x) = 9\sin(5x) + 3$.

a. $\frac{7\pi}{5}$

b. $\frac{4\pi}{5}$

c. $\frac{2\pi}{5}$

d. $\frac{\pi}{5}$

39 Find the period of $f(x) = 5\sin(5x) - 2$.

a. $\frac{7\pi}{5}$

b. $\frac{4\pi}{5}$

c. $\frac{\pi}{5}$

d. $\frac{2\pi}{5}$

40 Find the period of $f(x) = 5\sin(6x)$.

a. $\frac{\pi}{3}$

b. $\frac{\pi}{6}$

c. $\frac{4\pi}{3}$

d. $\frac{2\pi}{3}$

41 Find the period of $f(x) = 2\sin(4x) - 9$.

a. π

b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$

d. $\frac{3\pi}{2}$

42 Find the period of $f(x) = 5\cos(4x) + 1$.

a. π

b. $\frac{\pi}{4}$

c. $\frac{3\pi}{2}$

d. $\frac{\pi}{2}$

43 Find the period of $f(x) = 3\cos(4x) - 6$.

a. π

b. $\frac{\pi}{2}$

c. $\frac{3\pi}{2}$

d. $\frac{\pi}{4}$

44 Find the period of $f(x) = 5\sin(4x) - 2$.

a. $\frac{\pi}{4}$

b. π

c. $\frac{\pi}{2}$

d. $\frac{3\pi}{2}$

45 Find the period of $f(x) = 6\sin(8x) - 9$.

a. $\frac{5\pi}{4}$

b. $\frac{\pi}{2}$

c. $\frac{\pi}{8}$

d. $\frac{\pi}{4}$

46 Find the period of $f(x) = 3\cos(3x) + 8$.

a. $\frac{5\pi}{3}$

b. $\frac{\pi}{3}$

c. $\frac{2\pi}{3}$

d. $\frac{4\pi}{3}$

47 Find the period of $f(x) = 9\cos(3x) + 8$.

a. $\frac{5\pi}{3}$

b. $\frac{\pi}{3}$

c. $\frac{4\pi}{3}$

d. $\frac{2\pi}{3}$

48 Find the period of $f(x) = 3\sin(3x) + 1$.

a. $\frac{4\pi}{3}$

b. $\frac{\pi}{3}$

c. $\frac{5\pi}{3}$

d. $\frac{2\pi}{3}$

49 Find the period of $f(x) = 2\sin(2x) + 8$.

- a. $\frac{\pi}{2}$
- b. 2π
- c. $\frac{3\pi}{2}$
- d. π

Multiple choice — hard 15

50 Given $\sin \theta = \frac{20}{29}$ with θ in Quadrant I, find $\cos \theta$.

- a. $\frac{21}{29}$
- b. $\frac{20}{29}$
- c. $-\frac{21}{29}$
- d. $\frac{29}{21}$

51 Given $\sin \theta = \frac{5}{13}$ with θ in Quadrant I, find $\cos \theta$.

- a. $\frac{12}{13}$
- b. $-\frac{12}{13}$
- c. $\frac{13}{12}$
- d. $\frac{5}{13}$

52 Given $\sin \theta = -\frac{20}{29}$ with θ in Quadrant IV, find $\cos \theta$.

- a. $-\frac{20}{29}$
- b. $-\frac{21}{29}$
- c. $\frac{21}{29}$
- d. $\frac{29}{21}$

53 Given $\sin \theta = \frac{8}{17}$ with θ in Quadrant I, find $\cos \theta$.

a. $-\frac{15}{17}$

b. $\frac{8}{17}$

c. $\frac{15}{17}$

d. $\frac{17}{15}$

54 Given $\sin \theta = \frac{3}{5}$ with θ in Quadrant I, find $\cos \theta$.

a. $\frac{5}{4}$

b. $\frac{4}{5}$

c. $\frac{3}{5}$

d. $-\frac{4}{5}$

55 Solve $\sin \theta - \left(-\frac{\sqrt{3}}{2}\right) = 0$ for θ in $[0, 2\pi)$.

a. $\theta = \frac{4\pi}{3}$ or $\theta = \frac{5\pi}{3}$

b. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

c. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

d. $\theta = 0$ or $\theta = \pi$

56 Solve $2\cos \theta - 2\left(\frac{1}{2}\right) = 0$ for θ in $[0, 2\pi)$.

a. $\theta = \frac{\pi}{4}$ or $\theta = \frac{7\pi}{4}$

b. $\theta = \frac{3\pi}{4}$ or $\theta = \frac{5\pi}{4}$

c. $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$

d. $\theta = \frac{2\pi}{3}$ or $\theta = \frac{4\pi}{3}$

57 Solve $4\cos \theta - 4\left(-\frac{1}{2}\right) = 0$ for θ in $[0, 2\pi)$.

a. $\theta = \frac{\pi}{2}$ or $\theta = \frac{3\pi}{2}$

b. $\theta = \frac{2\pi}{3}$ or $\theta = \frac{4\pi}{3}$

c. $\theta = \frac{5\pi}{6}$ or $\theta = \frac{7\pi}{6}$

d. $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$

58 Solve $2\sin \theta - 2\left(\frac{\sqrt{3}}{2}\right) = 0$ for θ in $[0, 2\pi)$.

a. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

b. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

c. $\theta = \frac{4\pi}{3}$ or $\theta = \frac{5\pi}{3}$

d. $\theta = 0$ or $\theta = \pi$

59 Solve $\sin \theta - \left(-\frac{\sqrt{2}}{2}\right) = 0$ for θ in $[0, 2\pi)$.

a. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

b. $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

c. $\theta = \frac{5\pi}{4}$ or $\theta = \frac{7\pi}{4}$

d. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

60 A ladder leans against a wall, making a 52° angle with the ground. If the ladder is 100 ft long, find the height it reaches on the wall, to one decimal place.

a. 78.8

b. 76.8

c. 80.8

d. 83.8

- 61** A ladder leans against a wall, making a 70° angle with the ground. If the ladder is 79 ft long, find the height it reaches on the wall, to one decimal place.
- a. 76.2
 - b. 74.2
 - c. 79.2
 - d. 72.2
- 62** A ladder leans against a wall, making a 38° angle with the ground. If the ladder is 26 ft long, find the height it reaches on the wall, to one decimal place.
- a. 16.0
 - b. 14.0
 - c. 18.0
 - d. 21.0
- 63** A ladder leans against a wall, making a 50° angle with the ground. If the ladder is 92 ft long, find the height it reaches on the wall, to one decimal place.
- a. 75.5
 - b. 70.5
 - c. 72.5
 - d. 68.5
- 64** A ladder leans against a wall, making a 32° angle with the ground. If the ladder is 46 ft long, find the height it reaches on the wall, to one decimal place.
- a. 24.4
 - b. 26.4
 - c. 22.4
 - d. 29.4